



Environmental distribution of biofilm-forming antibiotic-resistant *Escherichia coli* associated with plastic surface materials

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Abstract

Plastic pollution is now an emerging issue worldwide, and the amount of plastic debris is rapidly increasing day by day in this decade. The surface of plastic contains a wide variety of biofilm-forming microorganisms that can pose a risk to human health. Studies showed that *Escherichia coli* is resistant to numerous classes of antibiotics; however, the prevalence of the bacterium on the environmental plastic surface is still unknown. The current study aimed at identifying biofilm-forming *E. coli* from the plastic surface collected from various environmental origins and determining the antibiotic-resistant pattern. A total of 90 plastic samples were collected from wastewater and open surface environments of Mymensingh Medical College, Bangladesh Agricultural University, and BCIC industrial areas of Mymensingh. Among these, 65 samples were found to be positive for the presence of *E. coli*. The plastic samples collected from drainage sources displayed the highest *E. coli* prevalence. By targeting the *malB* gene of the cultured samples, 36 *E. coli* isolates were positive out of 65, and the prevalence rate was 55.4%. There was a considerable variation in terms of the antibiotic-resistant pattern of the isolates. Randomly, 29 isolates were subjected to an antibiogram study. All of the isolates were resistant to imipenem (100%) and ceftazidime (100%), 79.40% were resistant to ampicillin, and 44.82% resistant to gentamicin. The beta-lactamase-producing genes *blaTEM* were detected in 51% (14/29) isolates that showed resistance to ampicillin. The biofilm-forming study revealed that 91.16% strong biofilm-forming *E. coli* isolates were resistant to ampicillin. Additionally, 18.18% non-biofilm-forming tetracycline-resistant *E. coli* isolates have been found in this study. In summary, to the best of our knowledge, this is the first study in Bangladesh to isolate and identify biofilm-forming antibiotic-resistant *E. coli* collected from environmental plastic surfaces, but further pathogenicity tests and resistome analysis are required to know the exact genetic resistance pattern.

Keywords *Escherichia coli* · Plastic surface · Biofilm · Antibiotic resistance

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Introduction

Escherichia coli, a rod-shaped gram-negative bacterium, belongs to the member of Enterobacteriaceae family, which is extensively studied in bacteriology. In proper condition, with enough amount of growth medium can multiply in about 20 min (Tenaillon et al. 2016). A lot of information about *E. coli*, including cultural and molecular characteristics, is available in the laboratory setting, but researchers are still looking for the ecological importance of it in environmental conditions. *E. coli* is considered a fecal indicator bacterium because it is widely used in water quality monitoring, and the survival and proliferation of the bacterium in the environment are now a public concern. In addition to this, although there are many antibiotics available, some serotypes and strains are causing infection in many countries; therefore, it is essential to discover the ecological relationship of these bacteria to spread to water and food (Jang et al. 2017).

Antimicrobial resistance (AMR) is now a major economic and public health-related problem in many areas of the world. *E. coli* is gradually becoming resistant to numerous antibiotic agents, and there is difficulty in terms of treatment due to resistance to most first-line antimicrobial treatments (Sabaté et al. 2008). In the present era, ampicillin (AMP), a semi-synthetic β -lactam antibiotic, is widely utilized to cure infections in livestock and people, though resistance to strains to these antibiotics has increased. 1–3 AMP is found to prevent bacterial cell walls at the time of the active replicating stage (Li et al. 2013).

Multi-drug resistant behavior is spreading rapidly in both animals and humans, and beta-lactam resistance is prevalent, which is the causative agent for illness (Parker et al. 2016). β -lactamases are encoded on the plasmid, and *E. coli*, an Enterobacteriaceae family bacteria, is the most common producer of this enzyme. Gradually, the bacterial community is going to be resistant to beta-lactam drugs, which cause widespread illnesses. *E. coli* is now also resistant to a wide range of antibiotics, for example, trimethoprim-sulfamethoxazole, aminoglycosides, and fluoroquinolones, which dramatically result in mortality and morbidity, deteriorate healthcare systems, and increase treatment costs. AmpC beta-lactamases (AmpC) and extended-spectrum beta-lactamases (ESBLs) are enzymes produced by plasmid-mediated beta-lactamases, and the absence of these biomolecules, resistance to beta-lactam antibiotics, is rising, that have a significant impact on the global health care sector (Pitout and Laupland 2008). The plasmid-mediated *AmpC* (CMY-2) is most often found in *E. coli* when it comes to animals and human host range (Alcalá et al. 2016).

Plastic waste has been considered a serious environmental issue globally. Wildlife species have been declining

every year due to this problem. Plastic is inserted into the food chain of higher trophic-level species when it is fragmented into small particles (< 5 mm) and ingested by biota. Due to the extensive use of plastic and rising plastic pollution, the harmful impact of microplastics and macroplastics on organisms and public health is an emerging concern. After entering the environment, plastic surfaces can colonize microorganisms rapidly. Fungi, microalgae, and bacteria are the most common microorganisms that are found on the plastic's surface in the environment (Amaral-Zettler et al. 2020; Wright et al. 2020). Biofilm formation on plastic surfaces and composition mechanisms has recently been studied (Rummel et al. 2017). In the last few years, biofilms formed by pathogenic microorganisms on plastic waste have been of environmental concern. The reason for this is that plastic waste has the ability to absorb biomolecules and organic matter, which is essential for the growth and survival of microorganisms. On the surface of plastic, microorganisms can go through horizontal gene transfer and can share antibiotic-resistant genes (Wu et al. 2019; Bowley et al. 2021). Many recent studies found viruses on the surface of plastic objects in schools, households, airports, and hospitals (Ikonen et al. 2018; Ong et al. 2020; Fong et al. 2020). People are not aware of plastic as a potential pathogen carrier and spreading disease, since plastic has a huge capacity for colonizing.

In this decade, researchers in many countries have been studying the significance of plastisphere communities, which helps us to understand the significance of plastisphere communities as potential human disease carriers (Rodrigues et al. 2019; Laverty et al. 2020; Galafassi et al. 2021). Only 12 infectious pathogens have been identified from 111 plastisphere studies, where three agents signify the existence of virulence-related genes (Jang et al. 2017; Beloe et al. 2022). Although methods are available, such as quantifying antimicrobial resistance genes and isolating fecal indicator organisms (FIOs), these cannot verify virulence, which is why culture-based methods allow researchers to isolate and identify potential infectious pathogens species and genus level (Wright et al. 2020). Moreover, functional protein is not always expressed from the presented gene; sometimes, extracellular DNA (eDNA) is a positive indicator for virulence genes that are released from dead cells (Masters et al. 1994; Li et al. 2019). With all extensive evidence several human pathogens can become “enriched” on plastics compared to natural materials, such as fine sands and decaying seaweed (Pham et al. 2021; Metcalf et al. 2022). Although some studies have been performed on plastic-associated bacteria, there is no study conducted yet about the association biofilm biofilm-forming *E. coli* with plastic in Bangladesh. Hence, it is essential to identify clinically important infectious agents found in environmental plastic debris that pose a

high health risk not only to the environment but also to public health.

Therefore, our study aimed at isolating and identifying *E. coli* from the surface of plastic material collected from wastewater and open surface environments and determining the biofilm-forming ability and antibiotic-resistant pattern of the isolated *E. coli*.

Methods

Sampling sites

The study was carried out at the Microbiology Laboratory at Bangladesh Agricultural University, and the samples were collected from Mymensingh Medical College, Bangladesh Agricultural University, as well as in various industrial areas in the Mymensingh Sadar region (Fig. 1). Plastic pollution in the City area of Mymensingh is gradually worsening and is now a major environmental concern for the government of Bangladesh. The drainage and open plastic surface are

populated with a wide range of bacteria, and day by day, they are becoming resistant to many antibiotic agents.

Sample collection and processing

The surface of plastic materials is a rich source of biofilm-forming bacteria. Our study aimed at isolating biofilm-forming *E. coli* from plastic sources. A total of 90 wet and dry (which is not associated with water) samples, including polybags, polyethylene, plastic bags, and tablet strips, were collected from drainage and open sources as represented in Table 1 and Supplementary Table 1. We selected this type of sample because limited studies have been conducted to determine the prevalence of biofilm-forming *E. coli* from these plastic sources. Additionally, by studying these samples, we can understand how plastic spreads harmful bacteria to the environment. To transfer each sample, a zip lock bag was used, and finally, all samples were delivered to the microbiology laboratory at Bangladesh Agricultural University. Initially, processing of the samples was done by cutting the plastic into small pieces and washing with PBS, and

Fig. 1 Study area map of sample collection. All of the samples have been collected from the Mymensingh Sadar region. The map was created by ArcMap.v.10.7 (ArcGIS Enterprise, ESRI, Redlands, CA, USA)

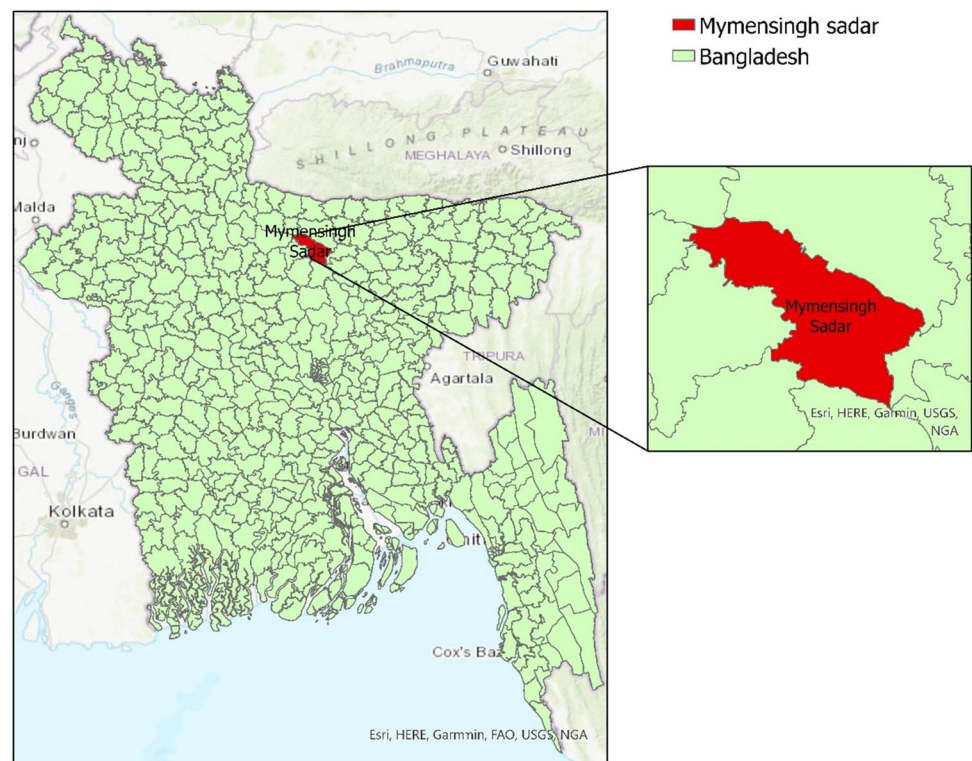


Table 1 Sample information of the study. All plastic samples were collected from various regions of Mymensingh Sadar

Samples sources	Bangladesh Agricultural University		Mymensingh Medical College		Industrial area	
	Dry plastic	Wet plastic	Dry plastic	Wet plastic	Dry plastic	Wet plastic
Drainage	6	8	11	10	5	12
Open	4	2	9	6	15	7
Sub total	10	10	20	16	15	19
Total	90					

then, homogenization was performed by adding distilled water. The humanized materials were placed into a sterile Eppendorf tube containing 5 mL of nutritional broth (HiMedia, India). The samples were then incubated aerobically overnight at 37 °C.

Isolation and molecular characterization of *E. coli*

To isolate *E. coli*, eosin methylene blue (EMB) agar (HiMedia, India) plates were prepared (Gazel et al. 2019). At first, the freshly overnight-grown broth cultures were streaked on the EMB agar plates with sterilized inoculating loops. Subsequently, the inoculated agar plates were incubated at 37 °C overnight to obtain pure colonies. For isolating pure *E. coli*, sub-cultures were done on EMB agar plates by using the isolates from the primary culture plates. The representation of single green-colored metallic sheen colonies growing on EMB agar plates indicates pure *E. coli* colony growth.

Genomic DNA extraction was performed by the boiling method (Yamagishi et al. 2016). A pure colony of *E. coli* was taken into 1000 µL of NB culture and incubated overnight. Initially, the overnight culture was centrifuged at 5000 rpm for 5 min, and the supernatant was discarded. Then, 1000 µL of PBS was added to the Eppendorf tubes and mixed by gentle vortexing. The mixture was again centrifuged at the same procedure, and the supernatant was discarded. After that, 250 µL of PBS was mixed into the Eppendorf tubes and boiled for 10 min. Then, the tube was immediately kept in ice powder for 10 min for cold shock, followed by centrifuging at 10,000 rpm for 10 min. Finally, the supernatant was collected as genomic DNA and stored at −20 °C for further use.

The PCR was performed with a final volume of 10 µL containing 1 µL nuclease-free water, 5 µL master mix (2X) (Promega, Madison, WI, USA), 1 µL of forward and reverse primer, and 2 µL of genomic DNA. All components were taken in a PCR tube and set in a thermocycler (Astec com. Ltd., Japan). Finally, PCR was run to detect genes involving bacterial virulence and antibiotic resistance (Table 2).

Biofilm-forming study of *E. coli* isolates

To evaluate the biofilm development of *E. coli*, phenotypically, a Congo red (CR) test was performed, as described previously (Luo et al. 2022). Initially, *E. coli* strains were cultured on Congo Red agar plates to examine biofilm formation. A total of 36 g saccharose (HiMedia, India) and 0.8 g CR (HiMedia, India) were mixed with 1000 mL of blood agar and incubated overnight at 37 °C. To check sterility, subsequently, *E. coli* culture was streaked on CRA plates and incubated at 37 °C for 48 h and 24 h. Biofilm formation of *E. coli* was observed based on their visible feature, whereas strong biofilm formers produce high levels of extracellular polysaccharides (EPS) that bind to Congo red dye, resulting a visible color change red to black; on the contrary, weak biofilm former do not have the ability to produce EPS, that is why color has not been changed. Dry filamentous crusty black, pink with a dark core, and smooth pink colonies indicate strong, intermediate, and non-biofilm forming, respectively.

Table 2 List of primers used to detect *E. coli* isolates, their virulence, and antibiotic-resistant genes

Determinants	Target genes	Primer sequence (5′–3′)	Amplicon size (bp)	References
Virulence	<i>malB</i>	F: GACCTCGGTTTAGTTCACAGA R: CACACGCTGACGCTGACCA	585	Wang et al. (1996)
Antibiotic resistance	<i>tetA</i>	F: GGTCACTCGAACGACGTCA R: CTGTCCGACAAGTTGCATGA	577	Randall et al. (2004)
	<i>blaTEM</i>	F: CATTTCGGTGTCCGCCCTTAT R: TCCATAGTTGCCTGACTCCC	793	Randall et al. (2004)

Antibiotic susceptibility test

Kirby-Bauer disk diffusion test was used to perform antibiotic susceptibility testing of all the isolated *E. coli* (Yin et al. 2021). Eleven commonly practiced antibiotic classes (ten antibiotics) in humans, animals, and aquacultures in Bangladesh were used. Isolated colonies of the same morphological type were selected, and the top of the colonies was touched with a loop, which was transferred into a tube containing 4–5 mL of nutrient broth. The broth cultures were then incubated at 37 °C for 6–7 h.

The turbidity of bacterial broth culture suspensions was adjusted with the 0.5 McFarland unit. The broth suspension was swabbed with a sterile cotton swab. To remove extra inoculums from the swab, it was spun several times and forcefully pushed against the inside wall of the tube above the fluid level. The dried surface of a Mueller Hinton agar plate (Hi Media, India) was infected by spreading the sterile agar surface with the swab. This method was done 3–4 times more, with the plate rotated 60° each time to ensure uniform dispersion of inoculum. Finally, the rim of the agar was swabbed. Before applying the antibiotic discs, the lid was left open for 3 to 5 min, but no more than 15 min, to allow any excess surface moisture to be absorbed.

The antibiotic discs were distributed over the inoculated agar plate's surface (Yuen et al. 2021). To establish thorough contact with the agar surface, each disc was pushed down. The discs were laid down separately, such that they were uniformly dispersed and no more than 24 mm apart from center to center. After applying the disc, the plates were inverted and put in an incubator for 15 min at 37 °C. The results were set out per standard guidelines of the Clinical and Laboratory Standards Institute (CLSI, 2024). Antibiotic susceptibility testing results were recorded as resistant, susceptible, and intermediate according to zone diameter interpretive standards provided by CLSI (2024). Multi-drug-resistant isolates were defined with resistance to three or more classes of antibiotics (Sweeney et al. 2022).

Statistical analysis

The entry of results was done using Excel 2016 spreadsheet (Microsoft Office 2016, Microsoft, Los Angeles, CA, USA) and the analysis was performed using the Statistical Package for Social Science- SPSS (IBM SPSS 25, IBM, Chicago, IL, USA) and GraphPad Prism version 8.4.3 (GraphPad Software, Inc.) (Ning et al. 2023).

The existence of any variations among the frequencies of the virulence genes among the prevalence of beta-lactamase genes was observed by the chi-square test for goodness-of-fit using IBM SPSS 25. The statistically significant *p*-value was less than 0.05 (*p*-value < 0.05).

Results

Molecular detection of *Escherichia coli*

The presence of the *malB* gene identifies *E. coli* species, and this locus is highly conserved in all strains. Randomly, we performed PCR by targeting the *malB* gene and found 36 positive samples out of 65. The visualization of the 585 amplicon size was represented under the ultraviolet transilluminator (Fig. 2).

Occurrence of *E. coli* isolates collected from various sources

Out of 90 samples, 65 (72.22%) samples were found for the growth of *E. coli* colonies on EMB agar plates. The positive isolates display green metallic sheen-colored, smooth, small, and round colonies. All colonies also represented favorable data on biochemical tests. Among all positive isolates, the plastic sample collected from MMCH showed the highest prevalence 75%, where 27 cultured positive *E. coli* were found out of 36. On the other hand, the sample collected from BAU and BCIC industrial areas displays 65% and 73% positive *E. coli* occurrence on EMB agar plates (Table 3).

Fig. 2 PCR amplification of the *malB* gene of *Escherichia coli*. N, negative control; M, 1 kb DNA marker; lane 1–8, *E. coli* isolates; PC, positive control

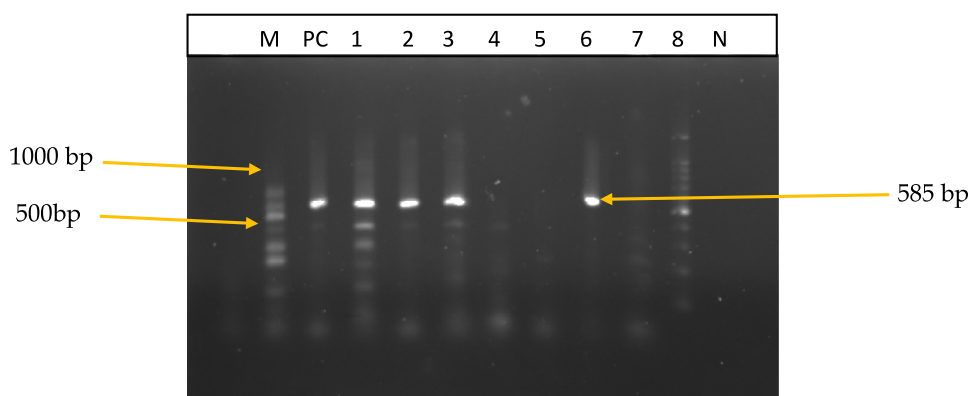


Table 3 Overall occurrence of plastic surface-associated *E. coli*

Area	No. of sample	Culture positive
BAU	20	13 (65%)
BCIC industrial zone	34	25 (73%)
MMCH	36	27 (75%)

Table 4 Occurrence of *E. coli* on various types of samples

Area	Open source	Drainage source
BAU	4/10 (40%)	9/10 (90%)
BCIC industrial zone	13/15 (86%)	12/19 (63%)
MMCH	12/20 (60%)	15/16 (93%)

We also studied the occurrence of *E. coli* based on the sample collection origin. Interestingly, plastic samples collected from drainage sources display the highest prevalence. In most cases, the drainage sample is filled with a wide range of bacteria. In terms of *E. coli*, more than 90% of samples were cultured positive on EMB plates. Drainage plastic samples from MMCH have a positive case of 93% (Table 4), and 90% positive *E. coli* isolates have been found in drainage plastic collated from BAU. In contrast, plastic from the open surface has limited *E. coli* occurrence, where 86% of cultured positive *E. coli* were isolated from the BCIC Industrial zone, followed by 60% from MMCH.

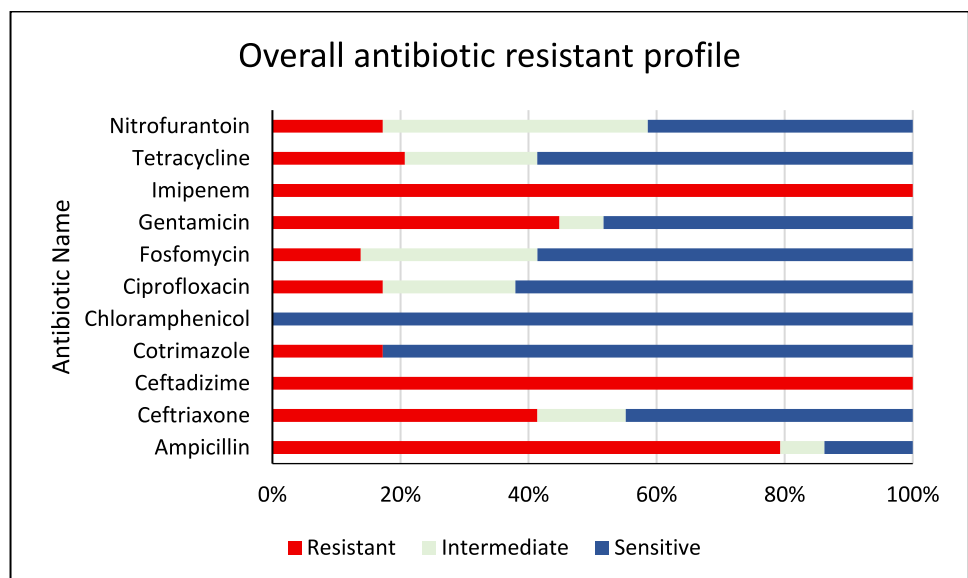
Antibiotic-resistant profile of selected *E. coli*

A total of 11 different antibiotic disks were used in this study to identify the resistance pattern of *E. coli*. Randomly, we selected 29 samples that were subjected to the antibiogram

study. In the selected samples, we found that Imipenem was the most resistant antibiotic for *E. coli*; the result of the experiment showed that 100% of isolates were resistant to Imipenem. In addition to this, ampicillin resistance was found in 79.31% which is considered the most important antibiotic agent for *E. coli*. Apart from these, ceftazidime was also identified as the highest resistant agent (100%) (Supplementary Table 3). Contrastingly, highly sensitive, intermediately sensitive isolates were exhibited against chloramphenicol, fosfomycin, nitrofurantoin, and clotrimazole. All isolates display sensitivity against chloramphenicol (100%), where the percentage of resistance and sensitivity of gentamycin and ceftriaxone against all isolates were almost 50/50. 48.27% of *E. coli* isolates were sensitive to gentamycin, followed by 44.82% resistant. Apparently, the resistant and intermediate sensitivity patterns of tetracycline were almost the same (20.68%) (Fig. 3).

Association of biofilm-forming *E. coli* with their antibiotic-resistant pattern

Biofilm formation was studied on Congo red agar, and the result was distributed into strong, intermediate, and non-biofilm formers. Among 29 isolates that were analyzed on the agar, 12 were found to be strong biofilm-forming. Additionally, 11 non-biofilm formers were displayed in the study, where only 6 isolates were observed as intermediate biofilm formers (Supplementary Table 2). We studied the association of biofilm formation with antibiotic resistance by utilizing 11 antibiotics of *E. coli*. Interestingly, 91.16% of strong biofilm-forming *E. coli* isolates were resistant to ampicillin. Tetracycline, which is one of the mostly used antibiotics common antibiotics that inhibits biofilm formation in *E. coli*, 18.18% non-biofilm forming, tetracycline-resistant *E. coli*

Fig. 3 Graphical presentation of antibiogram profile (overall) of *E. coli* isolated from BAU, MMCH, and BCIC industrial area

isolates have been found in our study (Table 5). The resistant pattern of imipenem, gentamycin, and nitrofurantoin was also observed in terms of strong biofilm formation. All strong biofilm formers are resistant to imipenem (100%), where the highest nitrofurantoin resistance was strong biofilm formers. On the other hand, the intermediate biofilm former exhibits the lowest resistance pattern. Resistant patterns of chloramphenicol and ceftazidime have no correlation with intermediate biofilm formation.

Prevalence of multi-drug resistant (MDR) *E. coli*

We also identified a multi-drug resistant (MDR) pattern of *E. coli* that resulted in 19 isolates that were MDR in nature out of 29. Overall, nine antibiotic-resistant patterns were observed among all isolates. The most prevalent resistance was considered as AMP, COT at 27%, followed by AMP, IMP at 18%. Among all isolates, three were found to be the most multi-drug antibiotic-resistant patterns (Table 6).

Molecular detection of the antibiotic-resistant gene of selected *E. coli* isolates

In the case of *Escherichia coli*, ampicillin and tetracycline are the common antibiotic agents. In our study, the highest possible ampicillin-resistant *E. coli* isolates have been identified by antibiogram strategy (79.31%), although tetracycline resistance was found in a limited case. Molecular detection of *tetA* and *blaTEM* genes, which are responsible for tetracycline and ampicillin resistance, has been confirmed by PCR protocol. Randomly, we selected ten isolates to detect the *blaTEM* gene, which revealed that around 40% of the samples are positive for the *blaTEM* gene (Fig. 4). In a similar way, 33.3% of positive isolates have been confirmed for *tetA* (Fig. 5).

Table 5 Association of antibiotic resistance patterns and biofilm formation in *E. coli* strains ($n=29$) detected in plastic samples collected from various areas

Virulence in different degrees of biofilm formation				
Categories	Antibiotics	No. (%) of strong biofilm former ($n=12$)	No. (%) of intermediate biofilm former ($n=6$)	No. (%) of non-biofilm former ($n=11$)
Phenotypic	Ciprofloxacin	0 (0%)	2 (33.33%)	3 (27.27%)
	Gentamicin	4 (33.33%)	4 (66.66%)	5 (45.45%)
	Ceftazidime	12 (0%)	6 (0%)	11 (0%)
	Tetracycline	1 (8.3%)	3 (50%)	2 (18.18%)
	Imipenem	12 (100%)	6 (100%)	11 (100%)
	Clotrimazole	1 (8.3%)	1 (16.66%)	3 (27.27%)
	Fosfomycin	3 (25%)	1 (16.66%)	0 (0%)
	Ampicillin	11 (91.16%)	5 (83.83%)	7 (63.63%)
	Chloramphenicol	0 (0 ^a)	0 (0%)	0 (0%)
	Nitrofurantoin	3 (25%)	1 (16.66%)	1 (9.09%)
	Cefotaxime	0 (0%)	2 (3.33%)	3 (27.27%)

Table 6 Phenotypic resistant pattern of selected *E. coli* isolated from plastic samples. Patterns 1, 2, 3, and 4 are MDR in nature

No. of pattern	Antibiotic resistance patterns	Number of antibiotics (classes)	No. of isolates	MAR index
1	AMP, TET, CIP, COT, IMP	5 (3)*	3	0.27
2	AMP, TET, COT	3 (2)	2	0.18
3	AMP, TET, CIP	3 (3)*	2	0.27
4	AMP, CIP, COT	3 (3)*	2	0.27
5	AMP, COT	2 (2)	4	0.18
6	AMP, FOS	2 (2)	4	0.18
7	AMP, IMP	2 (2)	23	0.18
8	AMP, TET	2 (2)	5	0.18
9	AMP, CIP	2 (2)	4	0.18
10	AMP	1 (1)	23	0.09

The asterisk (*) indicates MDR isolate

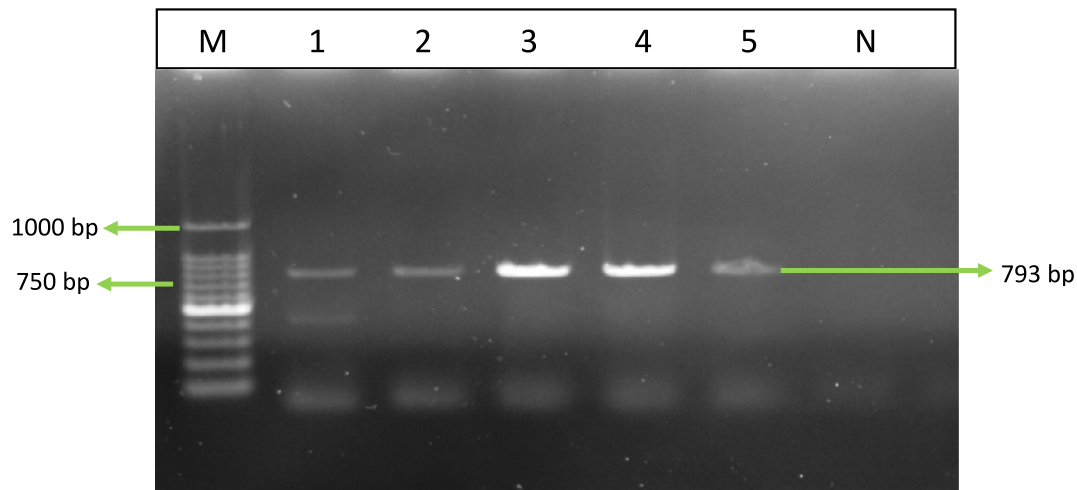
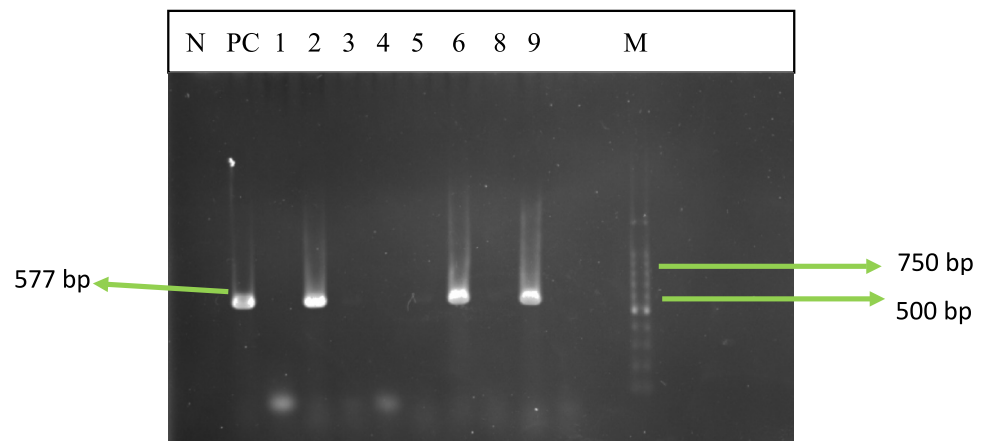


Fig. 4 PCR amplification of the *bla*_{TEM} gene of *Escherichia coli*. M, DNA marker; lane 2–5, representative *bla*_{TEM} gene of *E. coli* isolates; N, negative control

Fig. 5 PCR amplification of the *tetA* gene of *Escherichia coli*. M, DNA marker; lane 1–9, representative *tetA* gene of *E. coli* isolates; PC, positive control; N, negative control



Discussion

In the environment, soil and water are filled with a wide range of antibiotic-resistant bacteria, and in most cases, antibiotic resistance emerges from human, animal, and industrial wastes (Marathe et al. 2017). Wastewater from hospitals, either treated or untreated, considerably contributes to the growth of antibiotic-resistant bacteria, which is now a major health concern (Ali et al. 2017). In the current era, industries and medical centers are using numerous types of material, including plastic and polymers, but information regarding environmental risk factors is absent (Lanini et al. 2011). This waste can be hazardous and toxic, contain infectious agents to transmit diseases, and can result in public health risks (Blass et al. 2020).

Many harmful and disease-causing bacteria were found on plastic surfaces in the wastewater environment. Multidrug-resistant (MDR) bacteria with biofilm-forming

capability have huge environmental and public health risks (Jyoti et al. 2024). Antibiotic pollution not only disrupts the biodiversity of environmental bacteria but also changes the ecological equilibrium. Due to pollution and other reasons, bacteria are going to be resistant in the wastewater setting (Ahmad and Khan 2019; Bibi et al. 2023).

Plastic waste is a global issue threatening human health, animals, and the surrounding environment. In recent decades, plastic production has been rapidly increasing as well, and the amount of waste material is changing. It is predicted that 12,000 million metric tonnes of plastic waste will be present by 2050, despite recycling trends (Geyer et al. 2017; Geyer and Letcher 2020; Meral 2022; Bibi et al. 2023). In the environment, plastic waste results in ingestion and digestive blockages in animals and causes physical damage to coral reefs, providing rafts for invasive species and pathogens in the sea environment (Coe and Rogers 2012; de Stephanis et al. 2013). Plastic is converted into microplastic in the wastewater and open environment, enters the

food chain, and poses long-term effects on health and the ecosystem (Teuten et al. 2009; Tetu et al. 2019; Parvin and Tareq 2021).

On the surface of plastic, bacteria can easily develop biofilm formation, but plastic waste has been shown to inhibit the growth of oxygen-producing bacteria. The colonization of biofilm in the plastisphere is found in fecal indicator organisms such as *E. coli* and *Vibrio*. Therefore, the aim of the current study was to identify and detect biofilm-forming *Escherichia coli* from the surface of plastic and to detect the antibiotic-resistant pattern of the bacteria.

In this study, a total of 90 samples were collected from the open and wastewater environment from Mymensingh Medical College, Bangladesh Agricultural University, and BCIC industrial area of Mymensingh because these sites are the main areas that can produce a huge amount of plastic waste every day. Culturing of the sample was performed on EMB agar plates, which displayed 65 positive *E. coli* colonies. The prevalence rate was almost 72.22% which signifies that the plastic environment is the source of harmful bacteria. The highest prevalence rate was found in the sample collected from MMCH (75%); the reason for this might be due to the wastewater body. The medical hospital sites contained many water bodies that are responsible for colonizing microorganisms. The hard surface of waterborne plastic is a suitable environment for the formation of biofilm by opportunistic microbial colonizers (Teuten et al. 2009; Darcan 2012; Darcan and Aydin 2012). Furthermore, limited waste-associated plastic has been collected from the University region; thus, the bacterial prevalence rate is lower (65%). The molecular conformation of the isolates was done by PCR by targeting the *malB* gene. In a similar fashion, Zinnah et al. cultured and identified *E. coli* from various animal origins and found a 44% prevalence rate of the bacteria (Zinnah et al. 2007). Another study conducted by Nipa et al. collected raw vegetables and cultured them on EMB plates, and their result concluded a 40.64% prevalence rate (Nipa et al. 2011). In the current investigation, the *E. coli* isolates collected from the BAU area showed the lowest prevalence rate, despite the prevalence rate being higher compared to the study reported by Nipa et al. (2011).

The biofilm formation has a great contribution to the infection development by *E. coli*, and due to the presence of a tough biofilm layer, eradication of the bacteria is much more difficult (G et al. 2005). Biofilm provides a suitable environment for bacterial growth and has proper nutrient conditions. Some abiotic surfaces, such as plastic, glass, and medical implants, help to contribute to the attachment and biofilm formation of bacteria (Torres et al. 2002; Torres and Kaper 2003). We studied the biofilm-forming ability of some isolates and distributed them into strong, intermediate, and non-biofilm formers. About 41.37% of isolates exhibit strong biofilm-forming capability, which signifies

that environmental *E. coli* collected from wastewater and open environment may also be able to induce biofilm formation in the host body; this needs further investigation.

In recent ages, due to the excessive use of antibiotics in a haphazard way, *E. coli* have been developing multidrug resistance (MDR). MDR strains increase healthcare expenditures, morbidity, and death. It is found that MDR strains have increased dramatically in the last two decades. The main reasons for acquiring antibiotic-resistant *E. coli* are mutation, induction, and selective pressure (Sharma et al. 2016). Bacterial AMR gene transfer may be performed in vertical and horizontal ways.

To identify the resistant pattern, we used an antibiogram study of 29 *E. coli* isolates against 11 different antibiotics of 11 different classes. All selected isolates are resistant to imipenem and ceftazidime. Besides imipenem, ampicillin (79.4%) and gentamycin (44.82%) resistance were also found in the quarter of isolates. Some isolates exhibit intermediate sensitivity to ciprofloxacin (20.68%) and nitrofurantoin (41.37%). The sensitive nature of some isolates was identified in chloramphenicol and fosfomycin, whereas the resistant and sensitive proportions of gentamycin and cefotaxime were almost the same. Expansion of extended-spectrum-lactamase (ESBL), Enterobacteriaceae members raise resistance to cephalosporins (ceftriaxone, ceftazidime, and cefotaxime) (Sabaté et al. 2008). As well, resistance goes through the horizontal gene transfer process to survive a new environment, deliver antibiotic-resistance genes to other strains as well as other bacteria in the gastrointestinal system, and acquire resistance from other organisms (Sabaté et al. 2008). Similarly, Hassuna and her colleagues conducted a study on *E. coli*'s antibiotic resistance. They collected samples from children's urine and identified the biofilm-forming capacity of *E. coli* and found that cancer patient has a higher possibility of developing biofilm-forming antibiotic-resistant (Hassuna et al. 2024). Another study by Nipa et al. (2011) identified that ampicillin is the most abundant antibiotic for *E. coli* (Nipa et al. 2011). Faour-Klingbeil et al. (2016) isolated tetracycline-resistant *E. coli* (80%) (Faour-Klingbeil et al. 2016). In their study, more than two-thirds of MDR (69%) showed resistance to ampicillin and tetracycline, but the present study reported a higher level of resistance to ampicillin (79.44%). In addition, the PCR assay was conducted to detect resistant genes of *E. coli* isolates. The result found that beta-lactamase-producing gene *blaTEM* genes were observed in a total of 14 (51%).

We statistically correlate antibiotic-resistant pattern of the isolates with biofilm formation. Ampicillin, fosfomycin, and nitrofurantoin resistance were substantially higher in biofilm isolates compared to non-biofilm isolates. Similar work reported by Pompilio et al. discussed that strong biofilm-forming *E. coli* is more resistant to ceftazidime and ampicillin-resistant isolates did not show

biofilm-forming ability (Pompilio et al. 2018). Nitrofurantoin-resistant isolates have higher biofilm formation capability than was concluded by Hassuna et al. (2024). Plastic samples collected from MMCH showed the highest prevalence 75% *mal B* gene-based PCR, confirming 58% positive isolates. Antibiofilm study revealed that imipenem and ceftazidime resistance phenotype and ampicillin (79.4%) and gentamycin (44.82%) resistance were also found in the quarter of isolates. This is the first study in Bangladesh to isolate and identify biofilm-forming antibiotic-resistant *E. coli* collected from environmental plastic surfaces.

To develop infection and cause illness in healthy individuals, *E. coli* must be consumed in high quantities through the digestive tract. This microbe can grow in tissue by entering the small intestine, although some of them are destroyed in the stomach by the presence of acid. Gastric acid is the main molecule to destroys the majority of the eaten bacteria. *E. coli* strain O157:H7 can permit infection and make people ill, although the majority of the strains are benign. Initially, infection starts when a person takes contaminated food, as well as direct contact with infected persons and animals, and items can also transmit infection to healthy individuals. In Bangladesh, most people use polybags to carry food during shopping, which might be the reason for causing infection by *E. coli*. Diarrhea is the most common symptom of infection, which has serious consequences in small children and older people.

Conclusion

The present study was conducted to detect *Escherichia coli* from the plastic surface, identify biofilm-forming capability, and examine antibiotic-resistant patterns. Isolation and identification of *E. coli* were performed based on culture, staining, PCR, and biochemical tests. Out of all samples, 65 samples were found to be positive for *E. coli*. By targeting the *malB* gene of positive colonies found from the culture of 36 *E. coli* isolates, the prevalence rate was 56%. The biofilm-forming study revealed that 91.16% strong biofilm-forming *E. coli* isolates were resistant to ampicillin. In Bangladesh, this is the first examination to identify plastic surface-associated *E. coli*, but further findings, including studying the pathogenicity tests and the resistome analysis, could allow us to make some important findings on risk assessments, public health interventions, or environmental monitoring.

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Data availability The datasets supporting the conclusions of this study are included within the article (and its additional supplementary files).

Declarations

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